

VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH
MERN

Solution Architecture

Online Complaint Registration System — MERN Stack Complaint
Management Platform

Phase: Brainstorming & Ideation Phase

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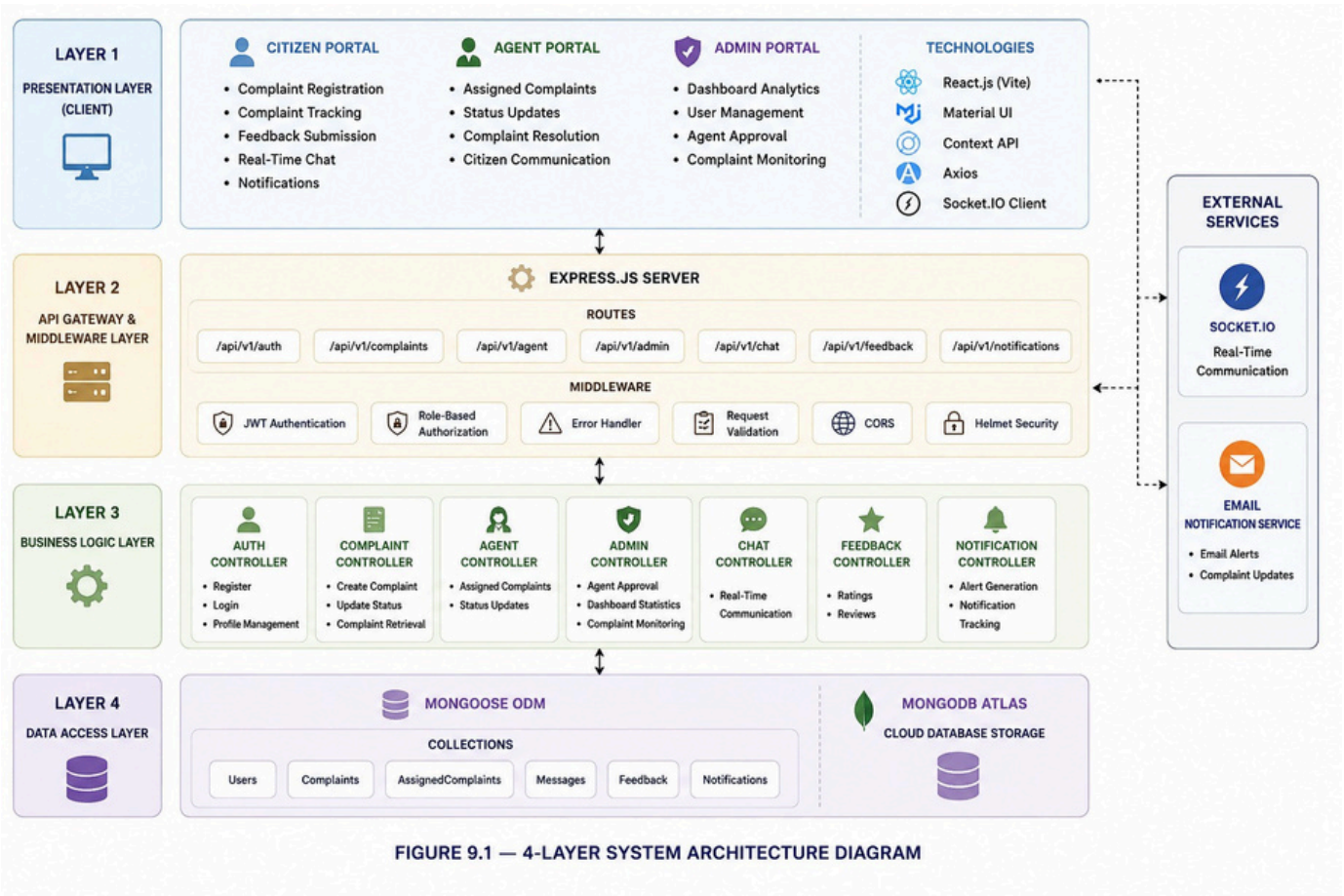
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Classification: Project Documentation — For Academic Submission

Solution Architecture

1. Layered System Architecture

The Online Complaint Registration System implements a layered architectural workflow boundary pattern to separate presentation components from back-end database schemas and WebSockets communication layers.



2. MVC Pattern Mapping

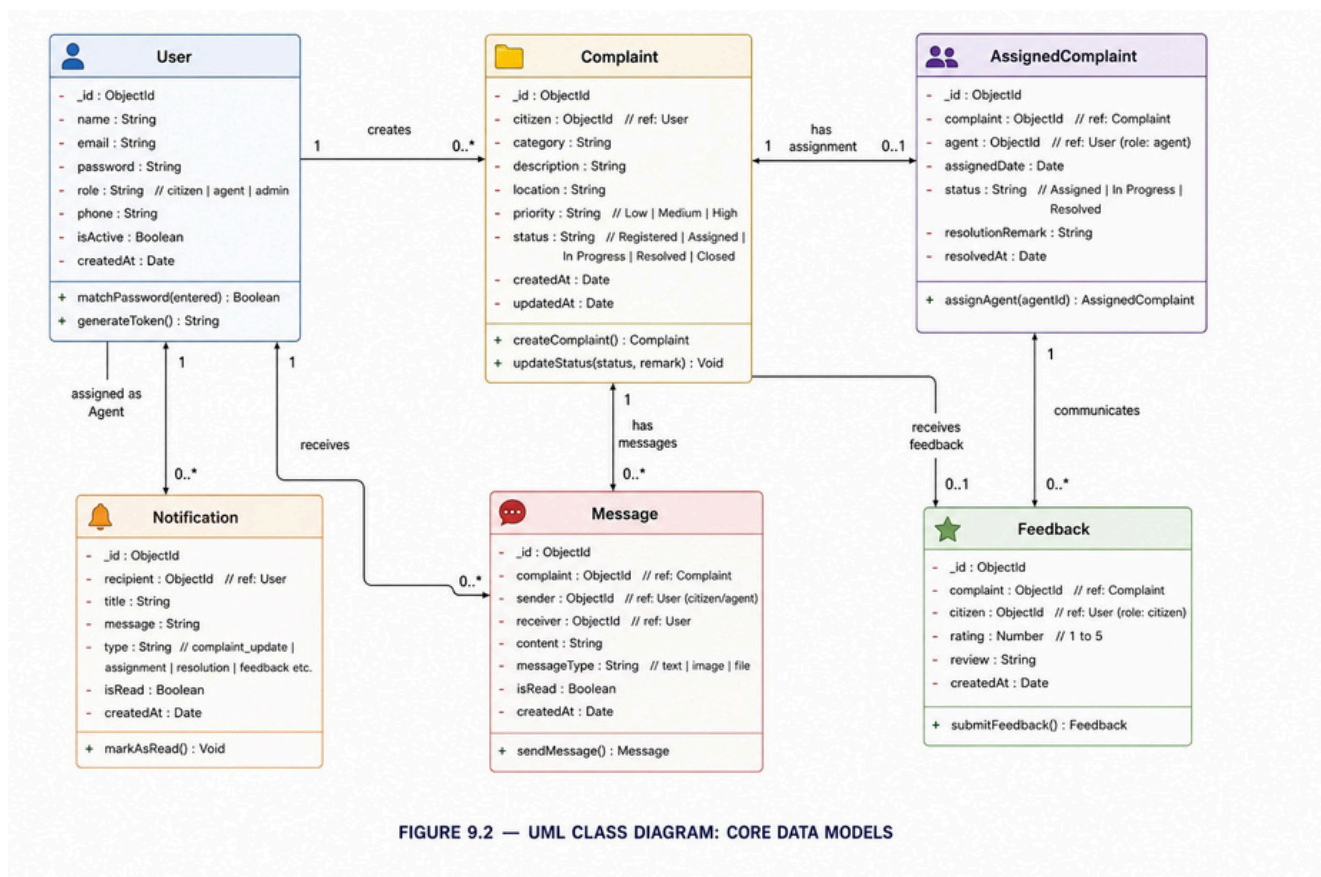
MVC Layer	MERN Component	Responsibilities
View	React.js Frontend SPA	Displays user interfaces, captures user interactions, communicates with APIs using Axios, manages component state and rendering.
Controller	Express.js Controllers	Receives requests, validates inputs, executes business logic, interacts with database models, generates API responses.

MVC Layer	MERN Component	Responsibilities
Model	Mongoose Models & MongoDB Atlas	Defines data structure, validation rules, and type constraints; executes schemas and models; interfaces with MongoDB Atlas.

The Model-View-Controller architecture splits computational tasks, allowing frontend visual components to change without causing logic defects in endpoint configurations.

3. UML Class Diagram – Mongoose Models

The data models define structural schemas, data types, and multiplicities regulating relationships across database entities.



4. Deployment Topology

The cloud deployment strategy isolates static resource distribution pipelines from transactional boundaries using standard server infrastructures.

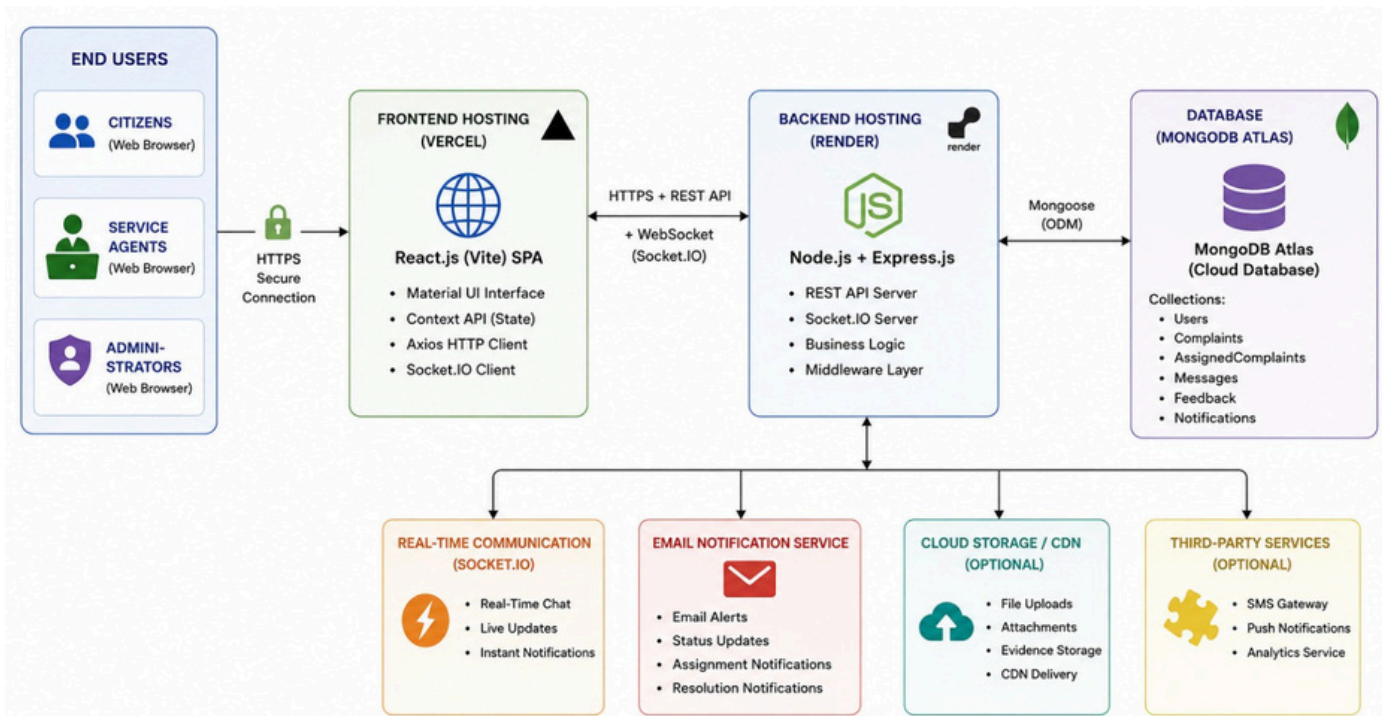


FIGURE 9.3 — CLOUD DEPLOYMENT TOPOLOGY

5. Deployment URL Reference

Component	Platform	URL
Frontend (React SPA)	Vercel	https://complaint-registration-frontend.vercel.app/
Backend (Node.js API)	Render	https://complaint-registration-api.onrender.com/
Database Layer	MongoDB Atlas	Cloud Hosted Cluster (Shared Sandbox Instance Tier)
Real-Time Communication	Socket.IO	Integrated Connection Protocol Layer with API Endpoint

6. Architectural Benefits

The layered configuration of the Online Complaint Registration System secures numerous technical benefits:

- **Scalability:** Independent hosting allocations on Vercel and Render allow independent performance vertical scaling.
- **Maintainability:** Separation of Concerns principles keep codebase layers decoupled and clearly auditable.
- **Security:** Restricting JWT tokens to HTTP-only client environments blocks cross-site script (XSS) leaks natively.
- **Real-Time Communication Support:** WebSockets eliminate HTTP pooling overhead on data state queries.

7. Conclusion

The Solution Architecture of the Online Complaint Registration System demonstrates a scalable, secure, and modular design capable of supporting complaint registration, complaint lifecycle management, real-time communication, notifications, analytics, and feedback collection. The separation of presentation, business logic, API management, and data access layers ensures maintainability, extensibility, and efficient system operation.
